

SIH1330: AI Multilingual Audio Tour Guide & Monument Recognition

Ministry of Tourism & Culture • Archaeological Survey of India (ASI)

Team: HackerMate Squad | Category: Software Edition

- Team Leader: Yash Shah (Full-Stack & Cloud Systems Architect)
- AI & Computer Vision Lead: Computer Vision Engineer (YOLOv8 & ResNet-50)
- Mobile App Developer: React Native / AR Camera Lead
- Speech & Vernacular Lead: BHASHINI API & Neural TTS Engineer
- UI/UX & Accessibility Lead: Female Teammate (Mandatory SIH Rule Compliant)
- QA & Pitch Presenter: Live Demo Video & ASI Documentation Lead

SLIDE 2: PROBLEM STATEMENT & PROPOSED SOLUTION

The Core Problem

1. Language Barriers: Tourists face communication gaps at heritage sites due to lack of multilingual guides.
2. Unverified Guide Scams: Visitors are overcharged by unregistered guides providing unverified history.
3. Artifact Identification Deficit: Museum visitors cannot identify specific temple carvings or museum artifacts.

Our Solution: SanskritiAI App

- Point smartphone camera at any monument/carving -> Instant Computer Vision Identification.
- Streaming Neural Audio Tour in 12+ Indian & International languages powered by BHASHINI AI.
- Spatial Geo-fencing & Indoor BLE Beacon triggers for seamless room-by-room audio playback.

SLIDE 3: TECHNICAL ARCHITECTURE & DATA FLOW

- Computer Vision Engine: Custom YOLOv8 + ResNet50 trained on 10,000+ ASI monument images.
- Speech & Translation: BHASHINI AI APIs + Neural Text-to-Speech (TTS) for natural vernacular voices.
- Spatial Positioning: GPS Geo-fencing + BLE Beacons for zero-latency room audio triggers.
- Cloud & DB Stack: FastAPI + PostgreSQL (PostGIS) on Supabase + SQLite for offline audio caching.

SLIDE 4: FEASIBILITY & 36-HOUR HACKATHON ROADMAP

- 00h - 12h: ASI Monument dataset curation, YOLOv8 fine-tuning, Supabase PostGIS schema.
- 12h - 24h: Mobile UI AR camera overlay, BHASHINI Vernacular Audio API integration.
- 24h - 32h: Offline SQLite audio caching, BLE beacon simulator, accessibility voice-over.
- 32h - 36h: End-to-end stress testing under low lighting, documentation & video pitch.

SLIDE 5: IMPACT, BENEFICIARIES & COMMERCIAL VIABILITY

- Impact Metrics: 90% cost reduction vs human guides, 98% artifact recognition accuracy, 12+ languages.
- Primary Beneficiaries: Ministry of Tourism, ASI, Domestic & Foreign Tourists, Visually Impaired Visitors.
- Commercial Model: B2G ASI White-Label Licensing + B2C In-App Day Pass (Rs 49 / \$1 per monument).

SLIDE 6: TEAM COMPOSITION & SIH COMPLIANCE

- Total Team Squad: 6 Members (Strict SIH Rule Satisfied)
- Female Representation: YES (Mandatory SIH Rule Satisfied)
- Role Diversity: Full-Stack Leader, Computer Vision Lead, Mobile Lead, Speech Lead, UI/UX Lead, QA Lead